3. Ohm's Law

The first lesson with an acceptance record

A meter-and-circuit verification chapter

Lesson 3 · two sessions · learner work: isolated two-AA source only

What you need

- Digital multimeter with intact leads and fused current input
- Switched holder and two AA cells
- 100, 220, 470 Ω, 1 kΩ, and 10 kΩ resistors
- Solderless breadboard and insulated jumpers
- Graph paper, pencil, and the meter manual

Predict first

At the same loaded source voltage, rank the five resistors by current. Predict the current for each using $I = V/R$, then mark which prediction is most sensitive to meter resolution.

THE DIAL, JACKS, AND LEADS FORM ONE INSTRUMENT

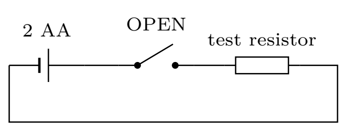
Current mode presents a low-resistance path. Never connect an ammeter across a cell or battery holder. Rewire only with the switch open and the holder disconnected. After every current reading, remove power and return the red lead to the V/Ω jack. Stop immediately for warmth, odor, damaged insulation, an unstable display, or an unexpected reading.

The questions the circuit must answer

1. Does the isolated resistor agree with its marked value?
2. Does a voltmeter placed across it report the loaded voltage?
3. Does an ammeter inserted in the only path report the path current?
4. Does V_R/I agree with resistance mode?
5. Are two independently rebuilt runs repeatable?
6. Does the V_R-against-I graph have the expected slope?
7. Is any disagreement larger than the combined uncertainty?
8. Can a fault be localized without random rewiring?

Connection atlas: copy the state, not merely the shape

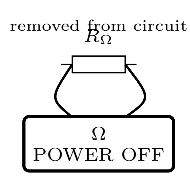
0. Safe starting state: open, disconnected, inspected



red lead parked in V/Ω; holder disconnected

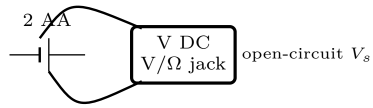
Read the resistor bands, inspect the leads, identify the fused current jack, and calculate maximum expected current before connecting cells.

1. Resistance: isolated part, no external power



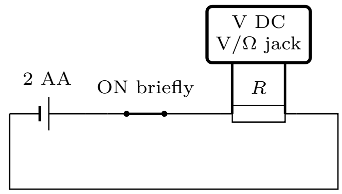
Short the probes together first and record their reading. A resistor still connected to other paths is not an isolated resistance test.

2. Source voltage: voltmeter across the source



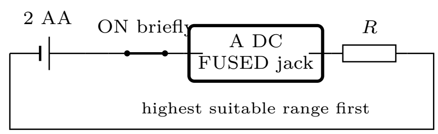
A voltmeter compares two nodes. Begin on a range above the source. Reverse probes if a negative sign appears; do not discard the sign silently.

3. Loaded voltage: voltmeter in parallel



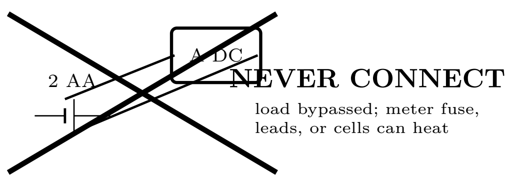
The resistor remains in the one current path; the voltmeter bridges only its two terminals. Record the source voltage under this load too.

4. Current: ammeter inserted in series



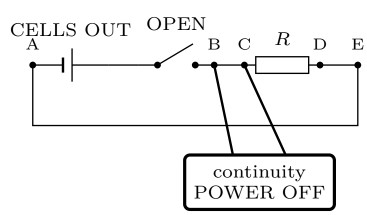
Break the loop and put the meter in the gap. All path current passes through it. Open, disconnect, and park the lead after the display settles.

5. Forbidden state: current input across the source



This is not a trial state. If it was approached or made, remove the holder and have the adult inspect the meter fuse and leads before reuse.

6. Continuity: cells removed, one segment at a time



A tone means only “below this meter’s threshold.” Record the actual resistance when contact quality matters.

State-transition check			
◯ switch open	◯ holder disconnected	◯ function/jacks checked	◯ expected range stated

Preflight record

Check	Observed	Required state	Adult initials
Lead insulation and probe guards		intact	
Current input marking and fuse rating		documented	
Probe-short resistance		recorded	
Open-circuit source voltage		below 3.4 V	
Largest predicted current		below chosen range/fuse	

Predict before measuring

Marked R	tolerance	assumed loaded V_s	predicted $I = V_s/R$	predicted $P = V_s^2/R$	ranking/reason
100 Ω					
220 Ω					
470 Ω					
1 kΩ					
10 kΩ					

Run the verification

1. Execute State 0. Do not proceed until maximum predicted current and meter range are compatible.
2. Execute State 1 for every resistor. Record function, range, resolution, and reading.
3. Build State 3 for the 10 kΩ resistor first. Record V_R and loaded V_s. Open and disconnect.
4. Rebuild as State 4. Record I. Open, disconnect, park the red lead.
5. Remove the resistor and rebuild the entire circuit for run 2. Repetition without rebuilding tests display stability; rebuilding also tests connection repeatability.
6. Continue toward smaller resistance only while the predicted current remains comfortably within the selected fused range.
7. Compute $R_{VI} = V_R/I$, $P = V_R I$, and percent difference from resistance mode. Plot V_R vertically against I horizontally.

Raw evidence — never erase an anomalous run

R marked	R_Ω	V_s loaded	V_R run 1/2	I run 1/2	R_{VI}	P	warm?	function/range/jack/resolution
100 Ω								
220 Ω								
470 Ω								
1 kΩ								
10 kΩ								

Worked result: arithmetic with units

For $V_R = 2.86$ V and $I = 12.9$ mA $= 0.0129$ A,

$$R_{VI} = \frac{2.86\text{ V}}{0.0129\text{ A}} = 222\ \Omega, \qquad P = (2.86\text{ V})(0.0129\text{ A}) = 0.0369\text{ W}.$$

For a 220 Ω, 5% part, the marked interval is $220(1 \pm 0.05) = 209$ to 231 Ω. The result is consistent with the marking, subject to meter uncertainty and run-to-run variation.

Uncertainty: say what the meter can support

Copy each relevant accuracy specification from the meter manual. For a specification $\pm(a\%\text{ reading} + b\text{ counts})$,

$$u_x = \frac{a}{100}|x| + bq,$$

where q is the displayed resolution. If $V = 2.86$ V, $q_V = 0.01$ V, and accuracy is $\pm(0.5\% + 2\text{ counts})$, then

$$u_V = 0.005(2.86) + 2(0.01) = 0.0343\text{ V}.$$

For a conservative resistance bound,

$$\frac{u_R}{R} \le \frac{u_V}{|V|} + \frac{u_I}{|I|}.$$

Add half the absolute difference between the two rebuilt runs if it is larger than the resolution contribution. Report no decimal place finer than u_R.

Uncertainty and agreement

Part	u_V	u_I	$R_{VI} \pm u_R$	marked interval	overlap and conclusion
100 Ω					
220 Ω					
470 Ω					
1 kΩ					
10 kΩ					

Questions from the graph and the circuit

1. Which axis choice makes slope equal resistance?
2. Why record loaded rather than only open-circuit source voltage?
3. Why is V_R/I a stronger check than current alone?
4. Should a best-fit line be forced through zero? What evidence decides?
5. What would curvature suggest about temperature or non-ohmic behavior?
6. Why might resistance mode and V_R/I differ?
7. For which part does lead resistance matter most?
8. For which current is meter resolution most important?
9. How can ammeter burden voltage change the circuit?
10. Why is a continuity tone not proof of a sound loaded connection?
11. When does "OL" mean open, and when only over-range?
12. Which anomalous result must stop the whole series?

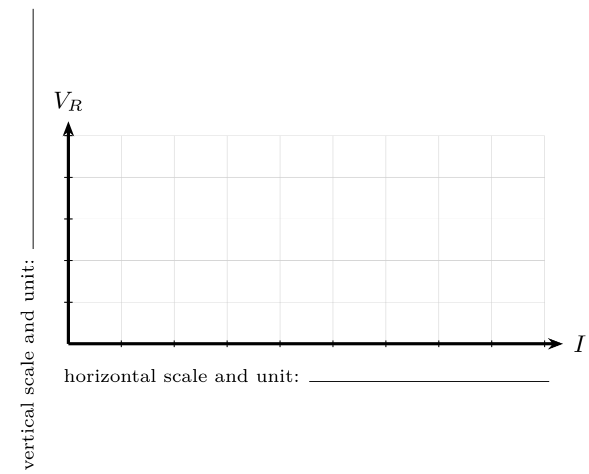
13. Does interval overlap prove the law? What does it actually show?

14. What independent observation would strengthen your conclusion?

Diagnose from evidence, not guesswork

Observation	De-energized checks	Bounded next measurement
No V_R, no I	open switch, empty/depleted holder, broken lead, loose row	verify V_s, then compare adjacent nodes A–E
V_R present, I reads zero	wrong jack/function/range, open meter lead, blown meter fuse	adult verifies meter on a known bounded path
Nearly full V_s across one jumper	open immediately downstream	remove cells; continuity-test only that segment
Near-zero V_R, unexpectedly high I	wrong resistor or bypass around it	remove cells; inspect and measure isolated R
Reading changes when a lead moves	loose/oxidized clip, damaged conductor, breadboard contact	replace one lead; rebuild both runs
Anything warms or smells	disconnect; quarantine warm/damaged hardware	no energized retest until cause is identified

Plot and test the model



Best-fit slope = __________ Intercept = __________ Largest residual = __________

Circle the point with the largest residual. Is its deviation explained by resolution, repeatability, source sag, contact resistance, or none of these? ____________________

Acceptance proof

NO ACCEPTANCE BY IMPRESSION

Criterion	pass?	Evidence
Only the isolated two-AA source was used	◯	
Every energized rewire began disconnected	◯	
Jacks, functions, ranges, and resolutions are recorded	◯	
Two rebuilt runs agree within the stated bound	◯	
R_Ω, R_{VI}, and marked intervals are compared	◯	
Graph has axes, units, points, fit, and slope	◯	
One fault path was reasoned through with cells removed	◯	
No heat, odor, damage, or unexplained anomaly remains	◯	
Red lead is parked in V/Ω; cells are removed	◯	

Decision: The measurements ◯ support ◯ do not support $V = IR$ for these parts over the tested range.

Claim defense

Current ranking evidence ______________________________

Resistance agreement evidence ______________________________

Graph evidence ______________________________

Largest anomaly and disposition ______________________________

One limitation of this proof ______________________________

Learner ______________ Adult inspection ______________ Date __________

Extension: change topology, repeat the proof

Predict the equivalent resistance and maximum current for two 470 Ω resistors first in series, then in parallel. Verify each network using the same connection atlas, two rebuilt runs, uncertainty bound, and acceptance gate. Do not use an LED as the Ohm's-law test article: it is nonlinear and polarity-sensitive.

> **Boundary retained.** This chapter authorizes only the isolated two-AA holder shown here. It does not authorize mains, opened appliances, large battery packs, power capacitors, ignition coils, sparks, or exposed high voltage. Adult supervision does not make a person electrically qualified.